

Analysis of Frequency Agile Radar using Block RIP

Shengyu Kang, Tianyao Huang*, Jie He

School of Computer & Communication Engineering
University of Science and Technology Beijing
Beijing, China

ksyer@xs.ustb.edu.cn,
{huangtianyao, hejie}@ustb.edu.cn

Tingting Wang

Qianyuan Laboratory
Hangzhou, China
xidian_wtt@126.com

Letian Guo

High Power Microwave Lab
Northwest Institute of Nuclear Technology
Xi'an, China
guoletian19901230@126.com

Abstract—Frequency agile radar (FAR) achieves extended target detection by synthesizing a wide effective bandwidth, and its range-Doppler reconstruction problem is formulated within the block compressed sensing (CS) framework. In this work, we analyze the block restricted isometry constant (RIC) of FAR's measurement matrices by exploiting their structured block form and leveraging tools from restricted isometry property (RIP) theory. We derive an upper bound on the block RIC that yields high-probability guarantees for the exact recovery of extended targets via block CS. The obtained bound is tighter than previous bounds derived via block coherence analysis and, up to logarithmic factors, has the same scaling as that of Gaussian sensing matrices, which is widely regarded as optimal and thus serves as a natural benchmark. This comparison highlights that the proposed block RIP analysis for FAR is essentially order-optimal, providing a rigorous theoretical foundation for block sparse range-Doppler reconstruction.

Index Terms—Frequency agile radar; compressed sensing; block restricted isometry property

I. INTRODUCTION

Frequency agile radar (FAR) is a pulse-based radar characterized by its ability to vary the carrier frequency over a wide bandwidth in a pseudo-random manner. This frequency-agility mechanism offers several critical advantages, including excellent extended target detection capabilities [1] and superior electronic counter-countermeasures (ECCM) performance [2].

This paper addresses the range-Doppler reconstruction problem for a FAR system. By synthesizing a wide bandwidth, FAR achieves high range resolution, which causes targets to occupy multiple range cells and manifest as extended targets. The target scene typically exhibits block sparsity, with each block representing the high-resolution range (HRR) profile of an extended target moving at a specific velocity. Despite the increase in bandwidth, the block sparsity of the target scene remains [1]. Consequently, block compressed sensing (CS) methods, such as $\ell_{2,1}$ -norm minimization, have demonstrated superior performance over conventional CS in FAR systems [1].

Performance analysis of block CS-based methods is crucial for guiding parameter selection and waveform design in FAR systems. Such analysis is categorized into three approaches. The first involves block coherence-based analysis. Wang et

al. [1] established sufficient conditions for exact block CS recovery via $\ell_{2,1}$ -norm minimization by analyzing the block coherence and spectral norm of the FAR's measurement matrix. However, the bounds based on block mutual coherence analyses are generally loose and exhibit intrinsic bottlenecks [3, Chapt. 6].

Another performance analysis approach is based on block restricted isometry property (RIP), which generally yields tighter recovery bounds than block mutual coherence analyses [4]. RIP-based analysis has been successfully applied to a variety of matrix ensembles, including Gaussian, Fourier, and other structured matrices [4]–[14]. However, block RIP analysis for FAR's measurement matrices has not been studied in prior work, and existing block RIP results cannot be directly applied to such matrices. Additionally, phase transition theory provides a further recovery bound, offering tighter predictions for Gaussian measurement matrices [15]. Li et al. [16] extended this theory to FAR's measurement matrices, providing empirical recovery guidelines. However, the application of phase transition theory in FAR lacks a rigorous theoretical foundation.

Building upon existing RIP research, we show that accurate reconstruction of the target scene is guaranteed with high probability, provided that the number of extended targets is on the order of $O(\frac{N}{M \ln M})$, where N is the number of pulses and M is the number of carrier frequency types in FAR. The obtained theoretical bound is tighter than the previous $O(\frac{N}{M \ln MN})$ in [1], which is based on an analysis of block coherence and the spectral norm of the measurement matrix. In principle, Gaussian matrices are optimal for sparse recovery [17], [18] and thus serve as a natural benchmark, and it is notable that our bound aligns with the Gaussian case for measurement matrices of the same shape, i.e., $O(\frac{N}{M})$, when the logarithmic factor is neglected [19]. Our theoretical results provide rigorous guarantees for CS algorithms in FAR systems.

The rest of the paper is organized as follows. Section II introduces the signal model of FAR. Section III presents preliminary knowledge of CS and RIP. Section IV presents the main conclusions of our study, with the proof details developed in its subsections. Section VI concludes the paper.

The following notation is used throughout this paper. Let $\mathbb{Z}^+$ and $\mathbb{C}$ denote the sets of positive integers and complex numbers, respectively. For $x \in \mathbb{C}$, $|x|$ represents the modulus

This work is supported by Fundamental Research Funds for the Central Universities (No. 00007802). Corresponding authors: Tianyao Huang.

of x . We use lowercase boldface letters to denote vectors (e.g., $\mathbf{a}$) and uppercase boldface letters to denote matrices (e.g., $\mathbf{A}$). Let $\mathbf{I}_n$ denote the $n \times n$ identity matrix, $n \in \mathbb{Z}^+$. The operator $(\cdot)^H$ represents the complex conjugate-transpose operator. For a vector $\mathbf{a}$, $[\mathbf{a}]_n$ denotes the n -th entry and $\|\mathbf{a}\|_i$ denotes the ℓ_i norm of $\mathbf{a}$, $i = 0, 1, 2$. For a matrix $\mathbf{A}$, $[\mathbf{A}]_{m,n}$ and $[\mathbf{A}]_i$ represent the (m, n) -th entry and the i -th column of $\mathbf{A}$, respectively. Finally, let $|S|$ denote the cardinality of a set S .

II. SIGNAL MODEL

In this section, we introduce the signal model of FAR as elucidated in [1], [20]. Consider a FAR system that transmits N pulses within one coherent processing interval (CPI) and hops among M carrier frequencies. For the n -th pulse, where $n \in \mathcal{N} = \{0, 1, \dots, N-1\}$, the carrier frequency is given by the following equation:

$$f_n = f_c + C_n \Delta f, \quad (1)$$

where f_c , C_n , and Δf represent the initial carrier frequency, the n -th randomized modulation code, and the frequency step interval, respectively, with $C_n \in \mathcal{M} = \{0, 1, \dots, M-1\}$. In this paper, we suppose that the number of pulses within a CPI is substantially large and exactly divisible by the number of carrier frequencies, and each modulation code is used P times uniformly without loss of generality, i.e.

$$P = N/M \in \mathbb{Z}^+. \quad (2)$$

The n -th transmitted pulse is given by

$$s_T(n, t) = \text{rect}\left(\frac{t - nT_r}{T_p}\right) e^{j2\pi f_n(t - nT_r)}, \quad (3)$$

where T_r and T_p denote the pulse repetition interval and pulse duration, respectively, with $T_r > T_p$, and $\text{rect}(\cdot)$ equals 1 for $x \in [0, 1)$ and 0 elsewhere. The transmitted pulse defined in (3) implies that the pulse is supported only over the specified duration T_p and repeats periodically with a period T_r .

Under the narrowband approximation [21], the sampled return in a fixed coarse range resolution (CRR) bin is modeled as a range-Doppler component. We discretize the range-Doppler plane onto a Cartesian grid with resolutions $1/M$ and $1/N$, respectively [22]. Assume that all scatterers are precisely located on the grid points. Consider a scatterer with range frequencies f_{r_p} , velocity frequency f_{v_q} , and complex coefficient $\gamma_{p,q}$. Then, the n -th sample is

$$y(n) = \sum_{p=0}^{M-1} \sum_{q=0}^{N-1} \gamma_{p,q} e^{j2\pi f_{r_p} C_n + j2\pi f_{v_q} n}. \quad (4)$$

Specifically, the components $f_{r_p} C_n$ and $f_{v_q} n$ in the phase explicitly capture the variations arising from the target's initial range (i.e., time delay) and velocity, respectively.

Our task is to perform range-Doppler reconstruction, i.e., to estimate $(\gamma_{p,q}, f_{r_p}, f_{v_q})$ from the sampled echoes $y(n)$. Denote the intensity vector $\mathbf{x} \in \mathbb{C}^{MN}$ as

$$[\mathbf{x}]_{p+Mq} = \gamma_{p,q}. \quad (5)$$

We now turn to define the measurement matrix $\mathbf{A}$. For each grid point $(\frac{p}{M}, \frac{q}{N})$, we construct a steering vector $\boldsymbol{\psi}_{p,q} \in \mathbb{C}^N$ as

$$[\boldsymbol{\psi}_{p,q}]_l = \frac{1}{\sqrt{N}} e^{j\frac{2\pi p}{M} C_l + j\frac{2\pi q}{N} l}, \quad (6)$$

where $\frac{1}{\sqrt{N}}$ is a normalization factor such that $\|\boldsymbol{\psi}_{p,q}\|_2^2 = 1$. For each Doppler index $n \in \mathcal{N}$, stack the M steering vectors to form

$$\boldsymbol{\Psi}_n = [\boldsymbol{\psi}_{0,n}, \boldsymbol{\psi}_{1,n}, \dots, \boldsymbol{\psi}_{M-1,n}] \in \mathbb{C}^{N \times M}. \quad (7)$$

Finally, the measurement matrix $\mathbf{A} \in \mathbb{C}^{N \times MN}$ is formed as

$$\mathbf{A} = [\boldsymbol{\Psi}_0, \boldsymbol{\Psi}_1, \dots, \boldsymbol{\Psi}_{N-1}]. \quad (8)$$

We now express (4) in matrix form, i.e.,

$$\mathbf{y} = \mathbf{A}\mathbf{x}, \quad (9)$$

where the vector $\mathbf{y} \in \mathbb{C}^N$ has entries defined as $[\mathbf{y}]_n = y(n)$.

It is worth noting that this paper focuses on analyzing the block RIP of the FAR's measurement matrix, which serves as a fundamental mathematical basis for radar target reconstruction performance. We acknowledge that our analysis relies on the strict assumption that all target and clutter scatterers are precisely located on the predefined grid—an assumption that rarely holds strictly in practical scenarios. While the resulting off-grid issue has been extensively studied in related literature [23]–[26], addressing this mismatch falls outside the scope of this work.

Equation (9) leads to a classical underdetermined linear inverse problem, which is ill-posed. Recall the definition of $\mathbf{x}$ in (5): targets typically occupy only a small subset of the range-Doppler grid, and extended targets span multiple adjacent range cells while sharing the same Doppler shift, which leads $\mathbf{x}$ to be a sparse or block sparse vector, and motivates solving (9) via CS and block CS algorithms. The success of such methods crucially depends on the structural properties of the measurement matrix $\mathbf{A}$, which are most conveniently characterized by the RIP and block RIP. Therefore, we next introduce these notions and the corresponding constants that quantify how well $\mathbf{A}$ preserves the geometry of sparse and block-sparse vectors.

III. PRELIMINARY KNOWLEDGE

A. Preliminary of CS and block CS

We now present the preliminaries of CS and block CS.

CS methods recover a sparse vector $\mathbf{x}$ from its lower-dimensional projections $\mathbf{y} = \mathbf{A}\mathbf{x}$ by finding the sparsest solution [27]. A widely adopted convex relaxation is the ℓ_1 -norm minimization problem, known as Basis Pursuit [28],

$$\min_{\mathbf{x}} \|\mathbf{x}\|_1 \quad \text{s.t.} \quad \mathbf{y} = \mathbf{A}\mathbf{x}. \quad (10)$$

The ℓ_1 -norm promotes sparsity by penalizing the magnitudes of individual coefficients. Under suitable RIP conditions on the measurement matrix, exact or stable recovery is guaranteed.

Now we turn to discuss block CS. As discussed in Section II, block sparsity arises in FAR when targets are extended in range. Each block of $\mathbf{x}$, as defined in (5), represents the HRR profile of an extended target moving at a specific velocity, thereby making $\mathbf{x}$ a block sparse vector. Similarly, convex relaxation is typically performed through the $\ell_{2,1}$ norm, the corresponding optimization problem is given in [29], [30]

$$\min_{\mathbf{x}} \sum_{i=0}^{N-1} \|\mathbf{x}_i\|_2 \quad \text{s.t.} \quad \mathbf{y} = \mathbf{A}\mathbf{x}. \quad (11)$$

Under block RIP conditions, such $\ell_{2,1}$ -norm minimization methods are guaranteed to recover the true block-sparse structure accurately in the noiseless case. By explicitly exploiting the block structure of $\mathbf{x}$, block CS methods typically provide improved recovery performance over standard CS in scenarios with extended targets [5], [16], [31].

The success of both CS and block CS depends on the suitable structural properties of the measurement matrix $\mathbf{A}$, which are commonly expressed in terms of the RIP and block RIP. These concepts are reviewed as follows.

B. Preliminary of RIP and block RIP

In this subsection, we review the RIP and block RIP through their associated restricted isometry constants (RICs), which provide precise quantitative measures of the quality of a measurement matrix and lead to tight bounds on sparsity levels [3, Chap. 6]. A matrix that satisfies the RIP is characterized by a small RIC, which is formally introduced as follows.

Definition 1. (RIC, [3, Def. 6.1]) *The K -th RIC $\delta = \delta_K(\mathbf{A})$ of a matrix $\mathbf{A} \in \mathbb{C}^{m \times N}$ is the smallest $\delta > 0$ such that*

$$(1 - \delta)\|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1 + \delta)\|\mathbf{x}\|_2^2, \quad (12)$$

for all K -sparse vectors $\mathbf{x} \in \mathbb{C}^N$.

The block RIC, which extends the standard RIC to block sparsity, often provides tighter recovery guarantees [4], and is formally defined as follows.

Definition 2. (Block RIC, [4]) *Let the matrix $\mathbf{A} = [\Psi_0, \Psi_1, \dots, \Psi_{N-1}] \in \mathbb{C}^{m \times NL}$ be formed by concatenating N blocks $\Psi_i \in \mathbb{C}^{m \times L}$, $i \in \mathcal{N}$. The K -th block RIC $\delta_b = \delta_{b,K}(\mathbf{A})$ of matrix $\mathbf{A}$ is the smallest $\delta_b > 0$ that satisfies*

$$(1 - \delta_b)\|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1 + \delta_b)\|\mathbf{x}\|_2^2, \quad (13)$$

for all K -block sparse vectors $\mathbf{x} \in \mathbb{C}^{NL}$.

In general, block RIP is easier to satisfy than standard RIP, and block CS algorithms can succeed under weaker conditions than those required for standard CS [4]. These results motivate our subsequent analysis of the block RIP of FAR's measurement matrix.

IV. BLOCK RIP ANALYSIS FOR FAR

In this section, we analyze the block RIP of FAR's measurement matrix $\mathbf{A}$ by exploiting its structured block form and deriving an upper bound on the block RIC. Due to space limitations, the detail proofs of some lemmas are remained in the full-length manuscript.

A. Main results

We first give the main result of this paper.

Theorem 3. *Consider the block-structured measurement matrix $\mathbf{A} \in \mathbb{C}^{N \times NM}$ of FAR as defined in (8). If, for any $\delta, \epsilon \in (0, 1)$, there exists an upper bound on the block sparsity level K , i.e.,*

$$K \leq \frac{\delta^2}{C} \frac{N}{M(\ln M - \ln \epsilon)}, \quad (14)$$

then the block RIC $\delta_{b,K}(\mathbf{A})$ is bounded by δ with probability at least $1 - \epsilon$. Here, the constant $C > 0$ is universal.

Proof. See Subsection IV-B. $\square$

Theorem 3 establishes that the block RIC of $\mathbf{A}$ is bounded from above with high probability, assuming a sufficiently large $M \gg \epsilon^{-1}$ and that the number of extended targets, or equivalently, the block number, aligns with the rational approximation $K = O\left(\frac{N}{M \ln M}\right)$.

To clearly illustrate the theoretical contribution of our block RIP analysis, Table I compares the proposed bound with the existing block coherence-based bound and the optimal Gaussian benchmark. As shown, our result provides a tighter guarantee for FAR measurement matrices and strictly approaches the optimal scaling when logarithmic factors are omitted.

TABLE I
COMPARISON OF THEORETICAL UPPER BOUNDS ON BLOCK SPARSITY K

Matrix type	Analysis method	Theoretical bound
FAR	Block RIP	$O\left(\frac{N}{M \ln M}\right)$
FAR	Block Coherence	$O\left(\frac{N}{M \ln(MN)}\right)$
Gaussian	Block RIP / Phase Transition	$O\left(\frac{N}{M}\right)$

B. Proof of Theorem 3

In this subsection, we prove Theorem 3 via the following steps. Initially, we establish that the block RIC of FAR's measurement matrix is bounded by the RIC of a set of partial Fourier matrices. By leveraging established RIC results for partial Fourier matrices and utilizing the union bound technique, we subsequently derive the bound on the block RIC of FAR's measurement matrix, which completes the proof.

We first introduce the set of pulse indices that share the same modulation code m , i.e.,

$$\Omega_m = \{n \mid C_n = m\}. \quad (15)$$

Suppose that each modulation code is used P times, i.e.,

$$P = |\Omega_m|. \quad (16)$$

Let $\mathbf{F} \in \mathbb{C}^{N \times N}$ denote the N -point discrete Fourier transform (DFT) matrix as $[\mathbf{F}]_{i,j} = e^{-j2\pi ij/N}$ [32]. The random partial Fourier matrix $\mathbf{F}_m \in \mathbb{C}^{N \times P}$ related to the random set Ω_m is defined as

$$\mathbf{F}_m = [\mathbf{F}]_{\Omega_m}. \quad (17)$$

We now show that the block RIC of $\mathbf{A}$ is bounded by the RIC of $\mathbf{F}_m$, as stated in the following proposition.

Proposition 4. *Consider $\mathbf{A}$ and $\mathbf{F}_m$ defined in (8) and (17), respectively. Then, the following inequality holds:*

$$\delta_{b,K}(\mathbf{A}) \leq \max_{m \in \mathcal{M}} \delta_K\left(\frac{1}{\sqrt{P}}\mathbf{F}_m^H\right). \quad (18)$$

Proof. See Subsection IV-C. $\square$

The term $\delta_K\left(\frac{1}{\sqrt{P}}\mathbf{F}_m^H\right)$, which appears on the right-hand side of (18), is bounded by the following lemma.

Lemma 5. *Consider $\mathbf{F}_m$ defined in (17). Let $\delta \in (0, 1)$. If the number of selected columns P satisfies*

$$K \leq \frac{\delta^2}{C} \frac{P}{\ln^4 N} \quad (19)$$

then the RIC satisfies $\delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right) \leq \delta$ with probability at least $1 - N^{-\ln^3 N}$. The constant $C > 0$ is universal.

Proof. See the full-length manuscript for the proof. $\square$

We reformulate Lemma 5 by making specific parameter substitutions. Let $\epsilon = MN^{-\ln^3 N}$, which leads to the probability bound being rewritten as $1 - N^{-\ln^3 N} = 1 - \epsilon/M$, and consequently, we obtain

$$\ln^4 N = \ln M - \ln \epsilon. \quad (20)$$

Next, by substituting (20) and (2) into (19), we derive

$$K \leq \frac{\delta^2}{C} \frac{P}{\ln^4 N} = \frac{\delta^2}{C} \frac{N}{M (\ln M - \ln \epsilon)}. \quad (21)$$

Consequently, when (21) holds, one has

$$\Pr \left(\delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right) \leq \delta \right) \geq 1 - \frac{\epsilon}{M}. \quad (22)$$

Applying the union bound [33] to (22) yields

$$\Pr \left(\max_{m \in \mathcal{M}} \delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right) \leq \delta \right) \geq 1 - \epsilon. \quad (23)$$

Combining Prop. 4 with (23) yields the proof of Theorem 3.

C. Proof of Proposition 4

In this subsection, we present the proof of Proposition 4. We first study the Gram matrices of certain submatrices of $\mathbf{A}$ in order to analyze its block RIC. Next, we show that the eigenvalues of these Gram matrices coincide with those of certain partial Fourier matrices. Finally, by exploiting these eigenvalue properties, we derive an upper bound on the block RIC of FAR's measurement matrix, thereby completing the proof.

Inspired by the definition of block RIC in Definition 2, we show an alternative formulation of the block RIC of $\mathbf{A}$ as follows:

$$\delta_{b,K}(\mathbf{A}) = \max_{\substack{\Lambda_b \subset \mathcal{N}, \\ |\Lambda_b| \leq K}} \sup_{\|\mathbf{x}\|_2=1} |\mathbf{x}^H \Psi_{\Lambda_b}^H \Psi_{\Lambda_b} \mathbf{x} - 1|, \quad (24)$$

where $\mathbf{x} \in \mathbb{C}^{|\Lambda_b|M}$ is an arbitrary unit vector, and $\Psi_{\Lambda_b} \in \mathbb{C}^{N \times |\Lambda_b|M}$ is defined as the juxtaposition of Ψ_n for $n \in \Lambda_b$.

The following lemma analyzes the eigenvalues of $\Psi_{\Lambda_b}^H \Psi_{\Lambda_b}$.

Lemma 6. *The eigenvalues of $\Psi_{\Lambda_b}^H \Psi_{\Lambda_b}$ are the union of those of $\frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H$ for $m \in \mathcal{M}$, where $\mathbf{V}_m^H = [\mathbf{F}_m^H]_{\Lambda_b} \in \mathbb{C}^{P \times |\Lambda_b|}$ and each m contributes $|\Lambda_b|$ of the total $|\Lambda_b| M$ eigenvalues.*

Proof outline. Rewrite $\mathbf{X}_{\Lambda_b} = \Psi_{\Lambda_b}^H \Psi_{\Lambda_b}$ as a block matrix whose (i, j) -th block $\Psi_{i,j} \in \mathbb{C}^{M \times M}$ is circulant, implying that $\mathbf{X}_{\Lambda_b}$ is a circulant block matrix. Its eigenvalues are given by the union of the eigenvalues of $\mathbf{C}_m \in \mathbb{C}^{|\Lambda_b| \times |\Lambda_b|}$, with one such matrix associated with each $m \in \mathcal{M}$ [34]. By element-wise comparison, we have $\mathbf{C}_m = \frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H$, which completes the proof. $\square$

This lemma implies that $\sup_{\|\mathbf{x}\|_2=1} |\mathbf{x}^H \Psi_{\Lambda_b}^H \Psi_{\Lambda_b} \mathbf{x} - 1|$ is expressed via $\frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H$ as

$$\sup_{\|\mathbf{x}\|_2=1} |\mathbf{x}^H \Psi_{\Lambda_b}^H \Psi_{\Lambda_b} \mathbf{x} - 1| = \sup_{\|\mathbf{v}\|_2=1} \max_{m \in \mathcal{M}} \left| \mathbf{v}^H \frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H \mathbf{v} - 1 \right|, \quad (25)$$

where $\mathbf{x} \in \mathbb{C}^{|\Lambda_b|M}$ and $\mathbf{v} \in \mathbb{C}^{|\Lambda_b|}$ are arbitrary unit vectors.

The following lemma shows that the supremum Rayleigh quotient of $\frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H$ is upper bounded by that of $\frac{1}{P} \mathbf{F}_m \mathbf{F}_m^H$ taken over $|\Lambda_b|$ -sparse vectors.

Lemma 7. *Consider an arbitrary matrix $\mathbf{U} \in \mathbb{C}^{N \times P}$ and its sub-matrix $\mathbf{V}^H = [\mathbf{U}^H]_{\Lambda_b} \in \mathbb{C}^{P \times |\Lambda_b|}$. Then, one has*

$$\sup_{\|\mathbf{v}\|_2=1} \left| \mathbf{v}^H \frac{1}{P} \mathbf{V} \mathbf{V}^H \mathbf{v} - 1 \right| \leq \sup_{\substack{\|\mathbf{u}\|_2=1, \\ \|\mathbf{u}\|_0 \leq |\Lambda_b|}} \left| \mathbf{u}^H \frac{1}{P} \mathbf{U} \mathbf{U}^H \mathbf{u} - 1 \right|,$$

where $\Lambda_b \subset \mathcal{N}$.

Proof. See the full-length manuscript for the proof. $\square$

Since $|\Lambda_b| \leq K$, we replace the constraint $\|\mathbf{u}\|_0 \leq |\Lambda_b|$ by $\|\mathbf{u}\|_0 \leq K$. Thus, Lemma 7 leads to

$$\sup_{\|\mathbf{v}\|_2=1} \left| \mathbf{v}^H \frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H \mathbf{v} - 1 \right| \leq \sup_{\substack{\|\mathbf{u}\|_2=1, \\ \|\mathbf{u}\|_0 \leq K}} \left| \mathbf{u}^H \frac{1}{P} \mathbf{F}_m \mathbf{F}_m^H \mathbf{u} - 1 \right|. \quad (26)$$

The right-hand side of (26) is written as $\delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right)$ for each fixed $m \in \mathcal{M}$, i.e.,

$$\delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right) = \sup_{\substack{\|\mathbf{u}\|_2=1, \\ \|\mathbf{u}\|_0 \leq K}} \left| \mathbf{u}^H \frac{1}{P} \mathbf{F}_m \mathbf{F}_m^H \mathbf{u} - 1 \right|. \quad (27)$$

Therefore, by substituting (25), (26) and (27) into (24), we derive

$$\delta_{b,K}(\mathbf{A}) \leq \max_{\substack{\Lambda_b \subset \mathcal{N}, \\ |\Lambda_b| \leq K}} \max_{m \in \mathcal{M}} \delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right) \stackrel{(a)}{=} \max_{m \in \mathcal{M}} \delta_K \left(\frac{1}{\sqrt{P}} \mathbf{F}_m^H \right),$$

where (a) stems from the independence of the matrix $\mathbf{F}_m$ relative to the support set Λ_b , which completes the proof.

V. SIMULATION

We numerically validate Lemma 6 via Monte Carlo simulations. The FAR system parameters are varied as follows: the number of pulses N from 128 to 512 (step 64), the number of carrier frequencies M from 8 to 16 (step 2, ensuring integer $P = N/M$), and the block sparsity K from 1 to 8. For each valid (N, M, K) combination, 50 independent random trials are conducted. In each trial, we compare the eigenvalues of the Gram matrix $\Psi_{\Lambda_b}^H \Psi_{\Lambda_b}$ against the concatenated eigenvalues of the M scaled sub-matrices $\frac{1}{P} \mathbf{V}_m \mathbf{V}_m^H$.

The ℓ_2 -norm of the difference averages 6.39×10^{-14} across all configurations. This machine-precision error exactly validates Lemma 6, demonstrating that the supremum Rayleigh quotient of the FAR's measurement matrix is equivalently evaluated via the partial Fourier matrices.

VI. CONCLUSION

In this work, we derive an upper bound on the block RIC of FAR's measurement matrix. Our results show that exact recovery of extended targets via block CS holds with high probability when the number of target blocks satisfies $K = O\left(\frac{N}{M \ln M}\right)$, which approaches the optimal Gaussian scaling when logarithmic factors are ignored. These results establish a rigorous inverse problem foundation for block CS in FAR signal processing and provide theoretically grounded guidance for system design and parameter selection.

ACKNOWLEDGMENT

We would like to thank Mr. Xingyu Xu for providing assistance with parts of the mathematical proofs.

REFERENCES

- [1] L. Wang, T. Huang, and Y. Liu, "Randomized Stepped Frequency Radars Exploiting Block Sparsity of Extended Targets: A Theoretical Analysis," *IEEE Transactions on Signal Processing*, vol. 69, pp. 1378–1393, 2021.
- [2] T. Huang, Y. Liu, X. Xu, Y. C. Eldar, and X. Wang, "Analysis of Frequency Agile Radar via Compressed Sensing," *IEEE Transactions on Signal Processing*, vol. 66, no. 23, pp. 6228–6240, Dec. 2018.
- [3] S. Foucart and H. Rauhut, *A Mathematical Introduction to Compressive Sensing*, ser. Applied and Numerical Harmonic Analysis. New York, NY: Springer New York, 2013.
- [4] Y. C. Eldar and M. Mishali, "Robust Recovery of Signals From a Structured Union of Subspaces," *IEEE Transactions on Information Theory*, vol. 55, no. 11, pp. 5302–5316, Nov. 2009.
- [5] Y. C. Eldar, P. Kuppinger, and H. Bolcskei, "Block-Sparse Signals: Uncertainty Relations and Efficient Recovery," *IEEE Transactions on Signal Processing*, vol. 58, no. 6, pp. 3042–3054, Jun. 2010.
- [6] R. G. Baraniuk, V. Cevher, M. F. Duarte, and C. Hegde, "Model-Based Compressive Sensing," *IEEE Transactions on Information Theory*, vol. 56, no. 4, pp. 1982–2001, Apr. 2010.
- [7] E. Elhamifar and R. Vidal, "Block-Sparse Recovery via Convex Optimization," *IEEE Transactions on Signal Processing*, vol. 60, no. 8, pp. 4094–4107, Aug. 2012.
- [8] Y. C. Eldar and H. Rauhut, "Average Case Analysis of Multichannel Sparse Recovery Using Convex Relaxation," *IEEE Transactions on Information Theory*, vol. 56, no. 1, pp. 505–519, Jan. 2010.
- [9] W. U. Bajwa, M. F. Duarte, and R. Calderbank, "Conditioning of Random Block Subdictionaries With Applications to Block-Sparse Recovery and Regression," *IEEE Transactions on Information Theory*, vol. 61, no. 7, pp. 4060–4079, Jul. 2015.
- [10] W. U. Bajwa, J. Haupt, G. Raz, and R. Nowak, "Compressed channel sensing," in *2008 42nd Annual Conference on Information Sciences and Systems*, 2008, pp. 5–10.
- [11] E. J. Candes and T. Tao, "Near-Optimal Signal Recovery From Random Projections: Universal Encoding Strategies?" *IEEE Transactions on Information Theory*, vol. 52, no. 12, pp. 5406–5425, Dec. 2006.
- [12] K. Jiang, S. Chen, and B. Tang, "The RIP and block-RIP analysis of Nyquist folding receiver for recovering signals," *EURASIP Journal on Advances in Signal Processing*, vol. 2016, no. 1, p. 92, Dec. 2016.
- [13] N. M. Gowda and A. P. Kannu, "Interferer identification in hetnets using compressive sensing framework," *IEEE Transactions on Communications*, vol. 61, no. 11, pp. 4780–4787, 2013.
- [14] S. T. Flammia, D. Gross, Y.-K. Liu, and J. Eisert, "Quantum tomography via compressed sensing: error bounds, sample complexity and efficient estimators," *New Journal of Physics*, vol. 14, no. 9, p. 095022, sep 2012.
- [15] D. Amelunxen, M. Lotz, M. B. McCoy, and J. A. Tropp, "Living on the edge: phase transitions in convex programs with random data," *Information and Inference*, vol. 3, no. 3, pp. 224–294, Sep. 2014.
- [16] Y. Li, T. Huang, X. Xu, Y. Liu, L. Wang, and Y. Eldar, "Phase Transitions in Frequency Agile Radar Using Compressed Sensing," *IEEE Transactions on Signal Processing*, vol. 69, pp. 4801–4818, 2021.
- [17] J. H. Lin and S. Li, "Block sparse recovery via mixed l_2/l_1 minimization," *Acta Mathematica Sinica, English Series*, vol. 29, no. 7, pp. 1401–1412, Jul. 2013.
- [18] Y. Wang, J. Wang, and Z. Xu, "Restricted p-isometry properties of nonconvex block-sparse compressed sensing," *Signal Processing*, vol. 104, pp. 188–196, 2014.
- [19] U. Ayaz, S. Dirksen, and H. Rauhut, "Uniform recovery of fusion frame structured sparse signals," *Applied and Computational Harmonic Analysis*, vol. 41, no. 2, pp. 341–361, 2016, sparse Representations with Applications in Imaging Science, Data Analysis, and Beyond, Part II.
- [20] T. Huang, Y. Liu, H. Meng, and X. Wang, "Cognitive random stepped frequency radar with sparse recovery," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 50, no. 2, pp. 858–870, Apr. 2014.
- [21] J. Costas, "A study of a class of detection waveforms having nearly ideal range—Doppler ambiguity properties," *Proceedings of the IEEE*, vol. 72, no. 8, pp. 996–1009, 1984.
- [22] M. A. Richards, J. A. Scheer, and W. A. Holm, *Principles of Modern Radar: Basic principles*. The Institution of Engineering and Technology, 2010.
- [23] Y. Chi, L. L. Scharf, A. Pezeshki, and A. R. Calderbank, "Sensitivity to Basis Mismatch in Compressed Sensing," *IEEE Transactions on Signal Processing*, vol. 59, no. 5, pp. 2182–2195, May 2011.
- [24] G. Tang, B. N. Bhaskar, P. Shah, and B. Recht, "Compressed Sensing Off the Grid," *IEEE Transactions on Information Theory*, vol. 59, no. 11, pp. 7465–7490, Nov. 2013.
- [25] Z. Yang, L. Xie, and C. Zhang, "Off-Grid Direction of Arrival Estimation Using Sparse Bayesian Inference," *IEEE Transactions on Signal Processing*, vol. 61, no. 1, pp. 38–43, Jan. 2013.
- [26] Y. Wang, M. Sznajder, and O. Camps, "A super-atomic norm minimization approach to identifying sparse dynamical graphical models," in *2016 American Control Conference (ACC)*. Boston, MA, USA: IEEE, Jul. 2016, pp. 1962–1967.
- [27] T. Huang, N. Shlezinger, X. Xu, D. Ma, Y. Liu, and Y. C. Eldar, "Theoretical Analysis of Multi-Carrier Agile Phased Array Radar," in *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*, vol. 2020-May. Institute of Electrical and Electronics Engineers Inc., 2020, pp. 4702 – 4706.
- [28] S. S. Chen, D. L. Donoho, and M. A. Saunders, "Atomic decomposition by basis pursuit," *SIAM Journal on Scientific Computing*, vol. 20, no. 1, pp. 33–61, 1998.
- [29] S. Bakin, "Adaptive regression and model selection in data mining problems," 1999.
- [30] M. Yuan and Y. Lin, "Model selection and estimation in regression with grouped variables," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 68, no. 1, pp. 49–67, 12 2005.
- [31] M. Mishali and Y. C. Eldar, "Reduce and boost: Recovering arbitrary sets of jointly sparse vectors," *IEEE Transactions on Signal Processing*, vol. 56, no. 10, pp. 4692–4702, 2008.
- [32] P. Duhamel and M. Vetterli, "Fast fourier transforms: A tutorial review and a state of the art," *Signal Processing*, vol. 19, no. 4, pp. 259–299, 1990.
- [33] G. Casella and R. L. Berger, *Statistical inference*, 2nd ed. Australia ; Pacific Grove, CA: Thomson Learning, 2002.
- [34] S. Rjasanow, "Effective algorithms with circulant-block matrices," *Linear Algebra and its Applications*, vol. 202, pp. 55–69, Apr. 1994.